

1. Sutherland 2.1
2. Sutherland 2.2
3. Sutherland 2.5 (AI-free)
4. Sutherland 2.7 as follows:

Suppose $\sim$ is an equivalence relation on a set X . Show that the equivalence classes partition X in the following sense. Let $\mathcal{C}$ denote the collection of equivalence classes. Then $\bigcup_{C \in \mathcal{C}} C = X$, and if $C, C' \in \mathcal{C}$ then either $C = C'$ or $C \cap C' = \emptyset$.

5. Give examples of collections of open intervals $I_i = (a_i, b_i)$ with $i \in \mathbb{N}$ such that

a) $\bigcap_{i \in \mathbb{N}} I_i = [0, 1]$

b) $\bigcap_{i \in \mathbb{N}} I_i = (0, 1)$

c) $\bigcap_{i \in \mathbb{N}} I_i = (0, 1]$

For part (a) give a formal proof. For the remainder, just give an example and a brief justification.

6. Sutherland 3.2 (No formal proof needed)
7. Sutherland 3.3
8. Sutherland 3.4
9. Sutherland 3.8 (Make your proof as short as possible)